msvtx_el1ecal

MC vs data · overview · rank 42 / 98

rank 42 / 98 by endcap D_equal

used by: NN1 barrel, NN1 endcap

rank-identical columns merged into this one:
    summedCaloClusters/el1ecal
    MS1Vtx_l1ecal_raw
    MS1Vtx_barrel_l1ecal
    MS1Vtx_endcap_l1ecal

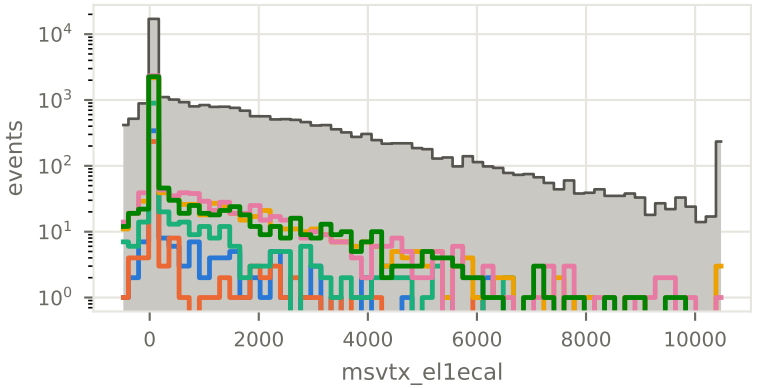
region	D_equal	noise	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg	rank
endcap	0.324	0.022	0.320	0.331	0.328	0.337	0.321	0.307	0.326	34,970	42/98
barrel	0.235	0.051	0.249	0.257	0.187	0.239	0.253	0.267	0.241	1,089	55/98
all	0.293	0.018	0.295	0.307	0.280	0.308	0.299	0.292	0.295	36,059	51/98

msvtx_el1ecal

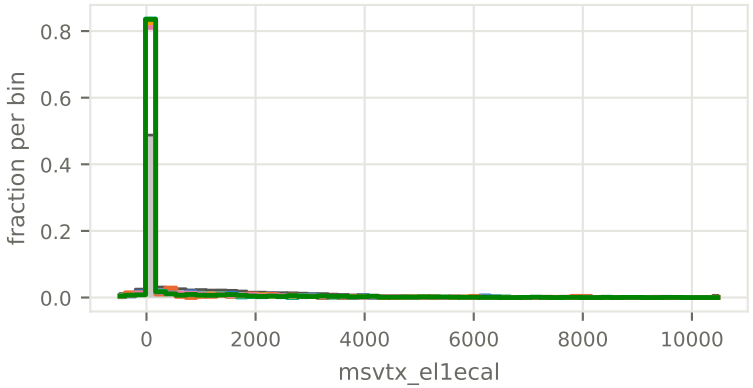
region: ENDCAP · rank 42 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.324 (noise 0.022) D_pooled 0.320 min 0.307 (mH600_mS150) max 0.337 (mH200_mS50) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

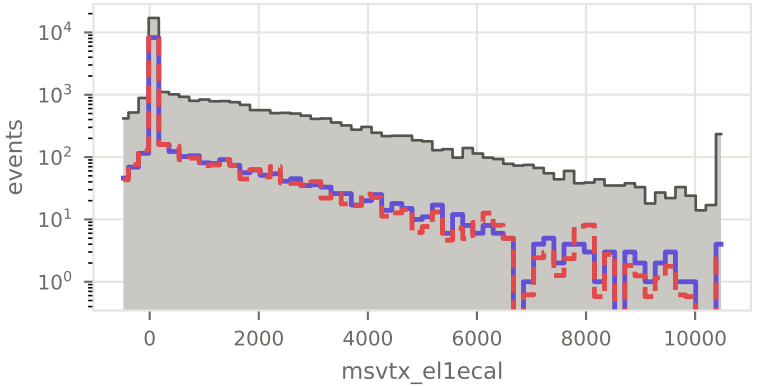


1b · each mass point vs background · normalized

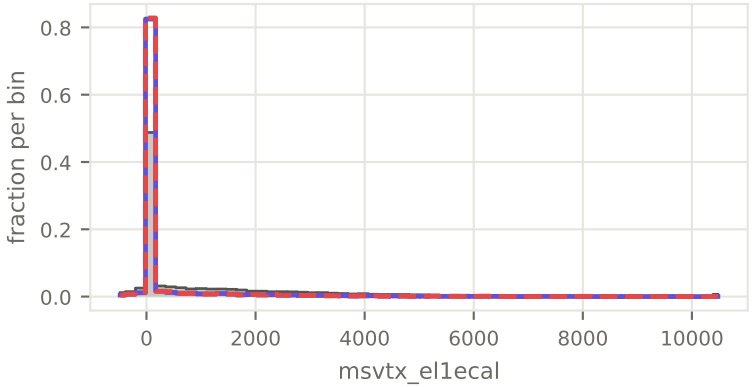


1 · each mass point
background n 34,970
mH125_mS15 D 0.331 n 411
mH125_mS55 D 0.328 n 281
mH200_mS50 D 0.337 n 1,071
mH400_mS100 D 0.321 n 2,648
mH600_mS150 D 0.307 n 2,882
mH1000_mS275 D 0.326 n 2,703

2a · combined signal vs background · raw

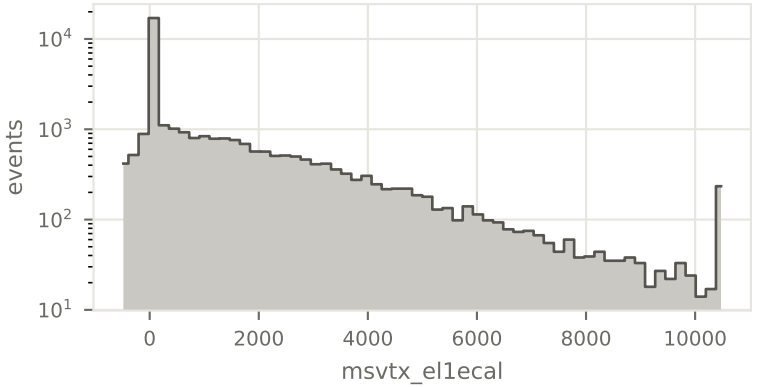


2b · combined signal vs background · normalized

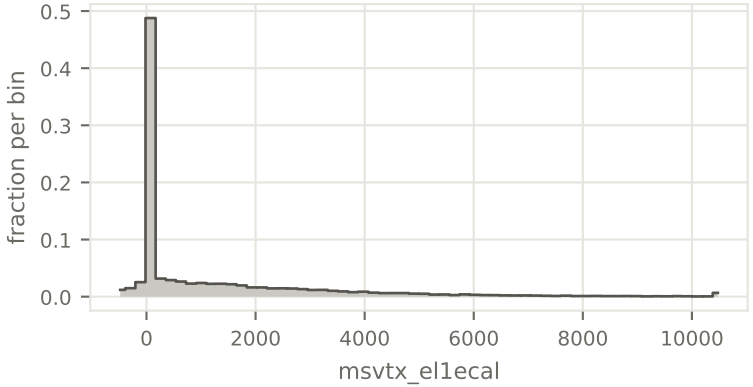


2 · combined signal
background n 34,970
combined, pooled D 0.320 n 9,996
equal weight per mass D 0.324

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

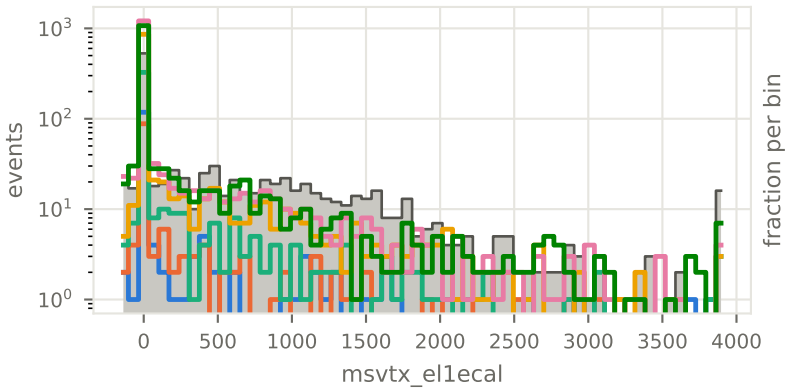
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_el1ecal

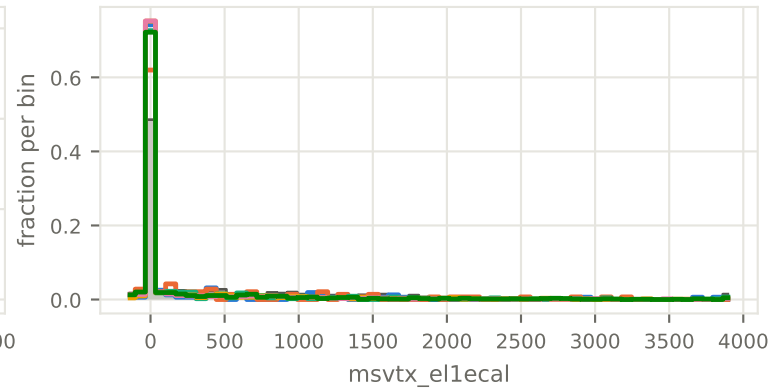
region: BARREL · rank 42 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.235 (noise 0.051) D_pooled 0.249 min 0.187 (mH125_mS55) max 0.267 (mH600_mS150) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

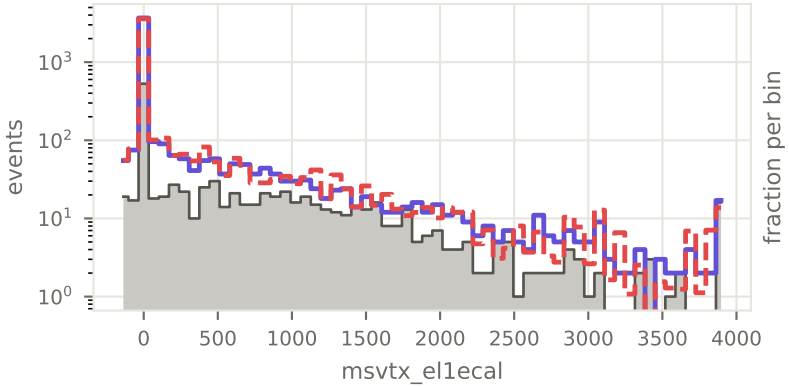


1b · each mass point vs background · normalized

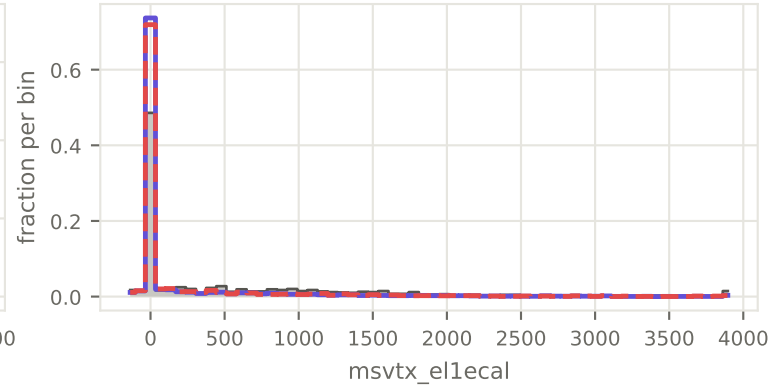


1 · each mass point
background n 1,089
mH125_mS15 D 0.257 n 159
mH125_mS55 D 0.187 n 142
mH200_mS50 D 0.239 n 449
mH400_mS100 D 0.253 n 1,144
mH600_mS150 D 0.267 n 1,608
mH1000_mS275 D 0.241 n 1,485

2a · combined signal vs background · raw

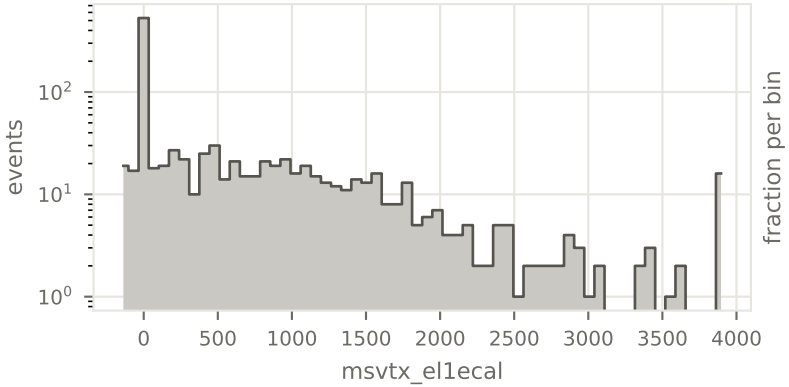


2b · combined signal vs background · normalized

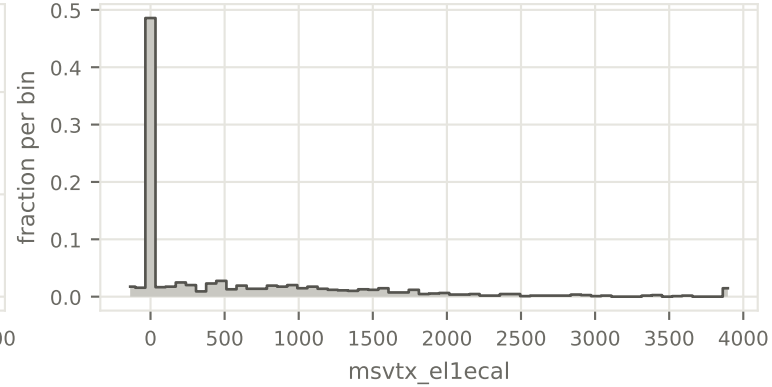


2 · combined signal
background n 1,089
combined, pooled D 0.249 n 4,987
equal weight per mass D 0.235

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

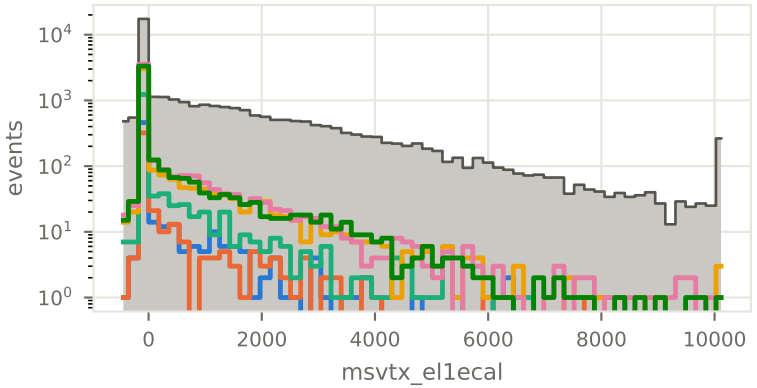
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_el1ecal

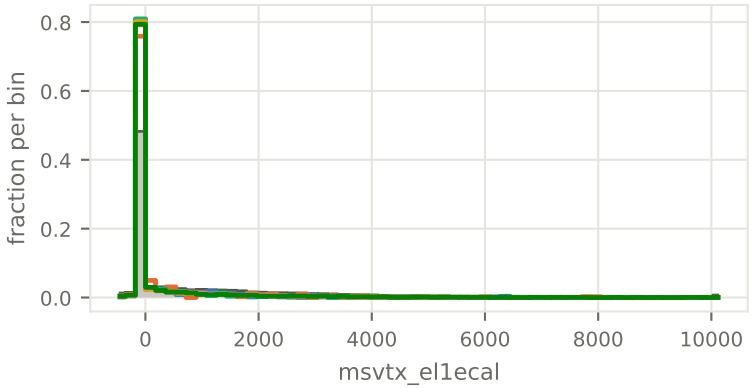
region: ALL · rank 42 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.293 (noise 0.018) D_pooled 0.295 min 0.280 (mH125_mS55) max 0.308 (mH200_mS50) bkg n = 36,059 sig n = 14,983

1a · each mass point vs background · raw

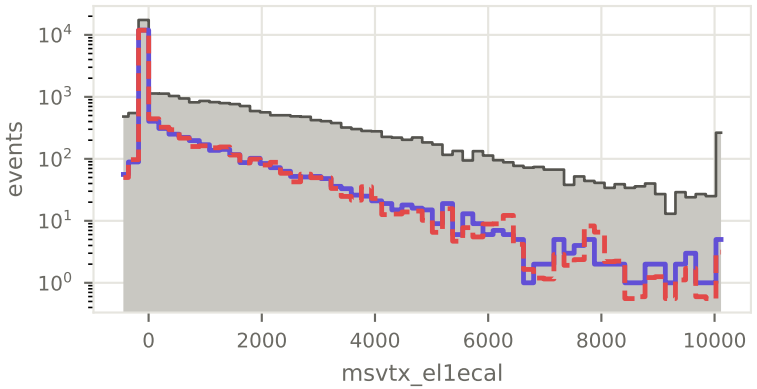


1b · each mass point vs background · normalized

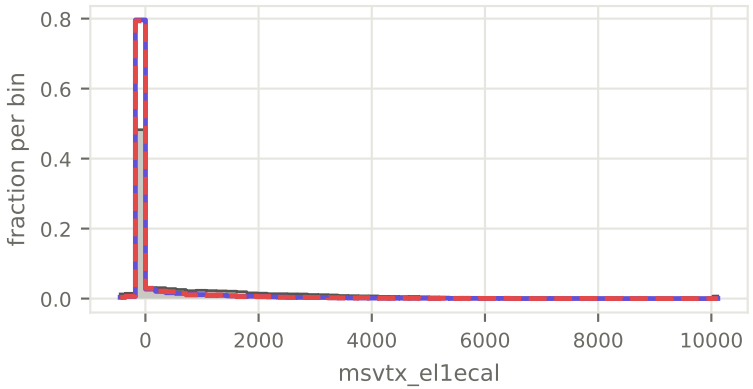


1 · each mass point
background n 36,059
mH125_mS15 D 0.307 n 570
mH125_mS55 D 0.280 n 423
mH200_mS50 D 0.308 n 1,520
mH400_mS100 D 0.299 n 3,792
mH600_mS150 D 0.292 n 4,490
mH1000_mS275 D 0.295 n 4,188

2a · combined signal vs background · raw

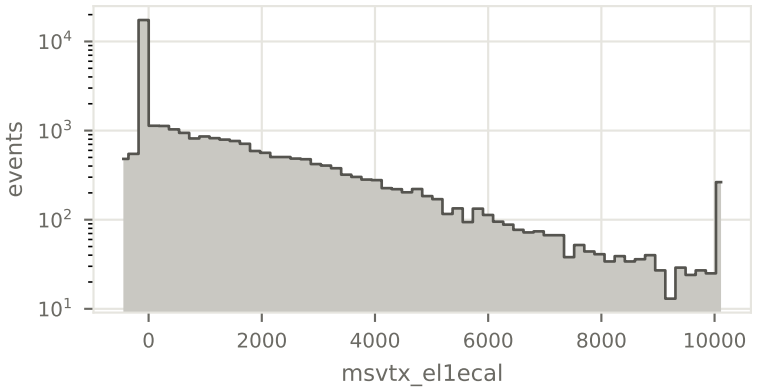


2b · combined signal vs background · normalized

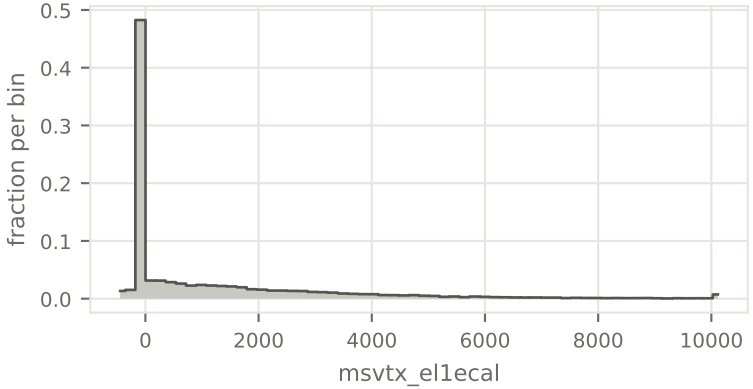


2 · combined signal
background n 36,059
combined, pooled D 0.295 n 14,983
equal weight per mass D 0.293

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 36,059

a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed